Net Zero Transition Plan - Financial & Emissions Analysis

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Role
Data Analyst / Sustainability & ESG Specialist
Project Duration
Q4 2024 - Q1 2025

PROJECT FOCUS
· Carbon Emissions Tracking & Analysis
· Abatement Strategy Evaluation
· Financial Impact Assessment
· Supply Chain Sustainability
· Net Zero Target Achievement

DATABASE & ARCHITECTURE
·Database Technology: SQLite + SQL
·Data Model: Normalized star schema with dimensional modeling
·Tables Created: 14 tables (10 core + 4 analytics tables)

TOOLS & TECHNOLOGIES
Data Warehouse: SQLite (normalized star schema)
SQL/Scripting: SQL (CREATE TABLE, INSERT, UPDATE, verification queries)
Business Intelligence: Power BI Desktop

EXECUTIVE SUMMARY - NET ZERO TRANSITION PLAN

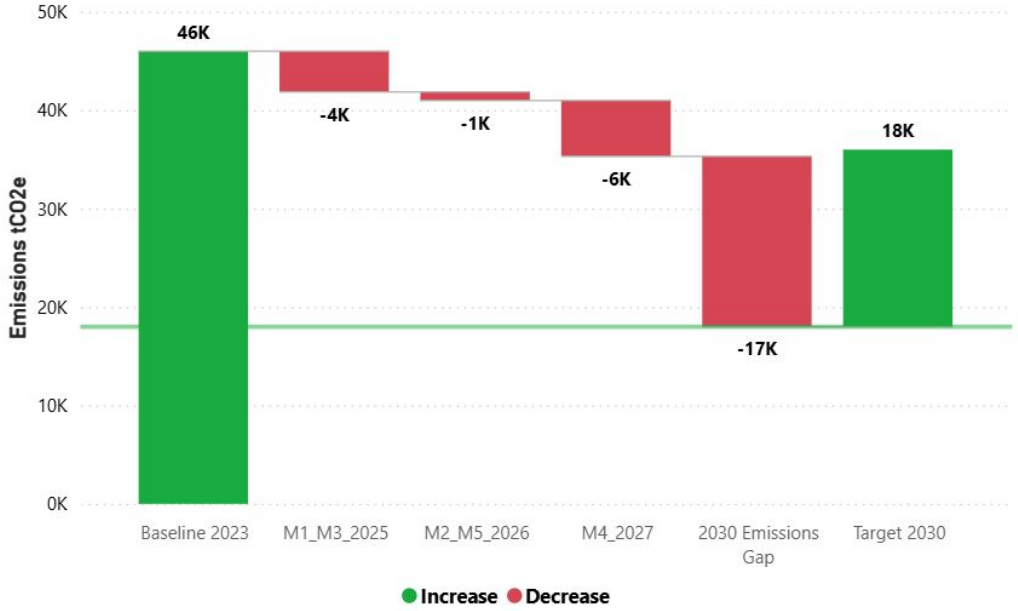
Baseline Emissions (2023)
45.805K

Emission Reduction Target 2030
18K

2030 Funding Gap
17.12K

← **CRITICAL** 69% Gap still unfunded

Path to 2030: Baseline Emissions → Planned Abatements → Remaining Gap to Target



Executive Summary: 2030 Emissions Pathway

Current Emissions (Baseline 2023): 45,805 tCO_2e
2030 Target: 18,000 tCO_2e
Gap to Target: 17,121 tCO_2e **(~69% of projected emissions still unabated)**
Key Implication: Significant additional action required by 2030 to close the gap. Options include:

- New / accelerated abatement measures
- Carbon removal or high-quality offsets
- Extension of timelines (with associated risks)

Financial Perspective:

- Planned investment: **$1.5M**
- Potential savings/value at $80/$tCO_2e$: **$2.72M** (from avoided costs or carbon pricing)

THE CORE PROBLEM - SCOPE 3 DEEP DIVE

Scope: Scope 1 | Scope 2 | Scope 3

Year: 2023 | 2025 | 2030 | 2035

Scenario: Baseline (2023) | Business as Usual

Scope 1: Direct Emissions

3,501 tCO2e

8% of Total

Scope 2: Electricity

3,444 tCO2e

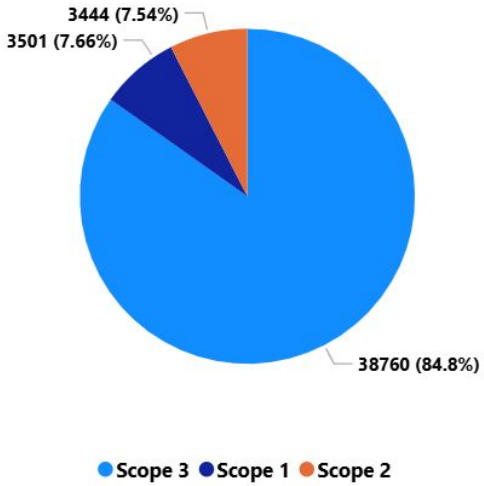
8% of Total

Scope 3: Supply Chain

38,860 tCO2e

85% of Total

2023 Emissions by Scope (tCO2e)



Scope 3 Breakdown

Category	Emissions tCO2e)	Perc
Business Travel	180	
Transport (Upstream)	780	
Purchased Goods & Services	37800	
Total	**38760**	

M4 (Supplier Engagement) Impact on Scope 3

Solves only 15% - 85% gap remains

Category	Phase	Value
M4 Reduction	Reduction	5670
Remaining Gap	After	33190
Scope 3 Total	Before	38860

SCOPE 3: THE CORE PROBLEM

- 85% of emissions from supply chain (38,860 tCO2e).
- M4 solves only 15% → 33,190 tCO2e gap remains (70% unfunded).

Additional strategies required:

Product redesign • Material substitution • Carbon offsets • Timeline extension

SCENARIO ANALYSIS

Scope: Scope 1 | Scope 2 | Scope 3

Year: 2023 | 2025 | 2030 | 2035

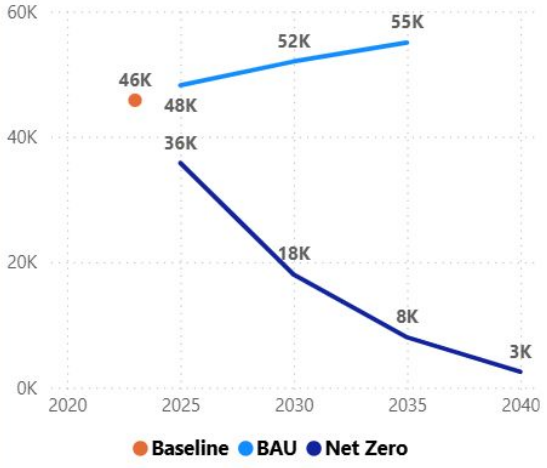
Scenario: Baseline (2023) | Business as Usual

Gap to close (BAU - NZ)

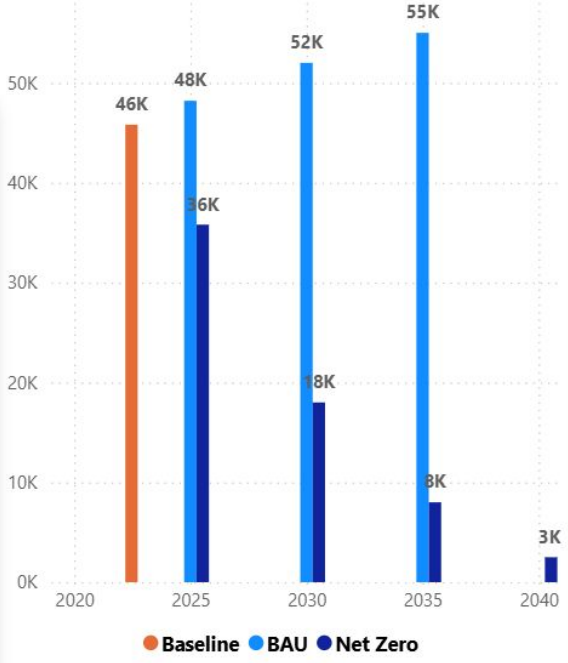
Emissions reduction needed

34,000 tCO2e

Emissions Trajectory: All Scenarios (2023-2040)



2030 Crossroads: Business as Usual vs Net Zero



SCENARIO ANALYSIS: THE CHOICE

2030 Baseline If Unchanged (BAU):
52,000 tCO2e emissions → **$4.16M** carbon cost 2030.

Net Zero Target:
18,000 tCO2e emissions → **$1.44M** carbon cost.

The Gap:
34,000 tCO2e reduction needed = **$2.72M** cost avoidance.

Decision Point:
Without intervention, emissions **GROW 13%** from 2023 baseline. With full Net Zero implementation, emissions **DROP 61%.** This is not an environmental choice - it's a **financial one**.

ABATEMENT MEASURES (MACC)

Scope: Scope 1 | Scope 2 | Scope 3

Year: 2023 | 2025 | 2030 | 2035

Scenario: Baseline (2023) | Business as Usual | Net Zero Pathway

Total Abatement Potential 2030
10,684 tCO2e

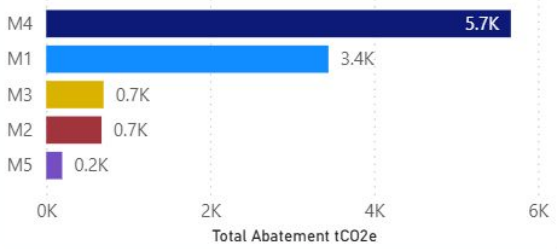
Total Capex
1.5M

Abatement Measures Financial Summary

Measure	Measure name	Scope	Abatement tCO2e 2030	Capex usd	Cost per tco2e usd	Payback years
M4	Supplier engagement	Scope 3	5670	200K	100.53	0.00
M1	Renewable electricity (PPA)	Scope 2	3444	300K	371.66	1.00
M3	Energy efficiency	Scope 1	700	200K	985.71	3.00
M2	Fleet electrification	Scope 1	675	600K	3,140.74	11.00
M5	Logistics optimization	Scope 3	195	200K	2,358.97	12.00
Total			**10684**	**1500K**	**6,957.61**	

Marginal Abatement Cost Curve (MACC)

Abatement by Measure



ABATEMENT MEASURES: THE STRATEGY

M4 (Supplier Engagement) is the leverage point:
- 5,670 tCO2e abatement = 53% of total reduction
- Requires earliest start (2024) for 2027 deployment

M1 (Renewable Energy) is the quick win:
- 3,444 tCO2e abatement = 32% of total reduction
- Deploys 2025, lowest execution risk
- Eliminates 100% of Scope 2 emissions

M3 (Energy Efficiency) has best cost efficiency:
- 700 tCO2e abatement = 7% of total reduction
- $285/tCO2e (lowest cost per unit)
- Deploys 2025, quickest implementation

M2 (Fleet Electrification) is capital intensive:
- 675 tCO2e abatement = 6% of total reduction
- $888/tCO2e investment required
- ROI: -91% | Payback: 11.1 years

M5 (Logistics Optimization) has limited impact:
- 195 tCO2e abatement = 2% of total reduction
- Only addresses 2% of Scope 3
- Supports M4 supplier engagement strategy

COMBINED PORTFOLIO:
- Total abatement: **10,684 tCO2e** (23% of baseline)
- Total CAPEX investment: **$1.5M**
- Annual carbon cost avoidance: **$2.72M**
- Portfolio **ROI: 81.3% | Payback: 0.55 years** (6.6 months)

FINANCIAL IMPACT

Scope: Scope 1 | Scope 2 | Scope 3

Year: 2023 | 2025 | 2030 | 2035

Scenario: Baseline (2023) | Business as Usual

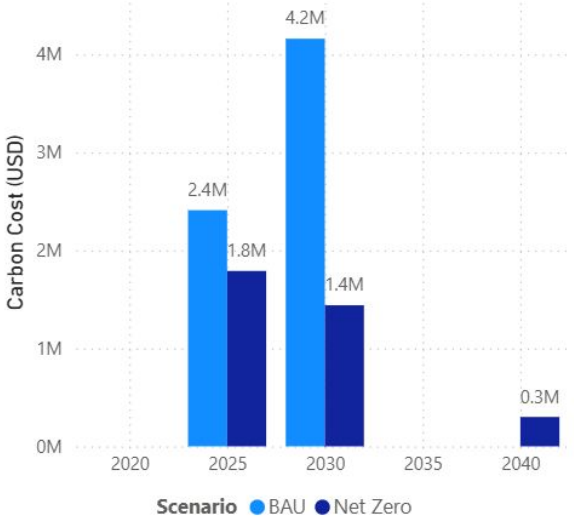
Annual Carbon Cost Avoided
$2,720,000

Total Investment Required
$3,350,000

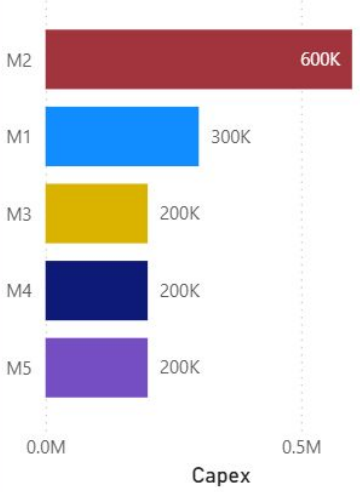
Payback Period
0.55 years

Annual ROI
81.3%

Annual Carbon Cost: BAU vs Net Zero



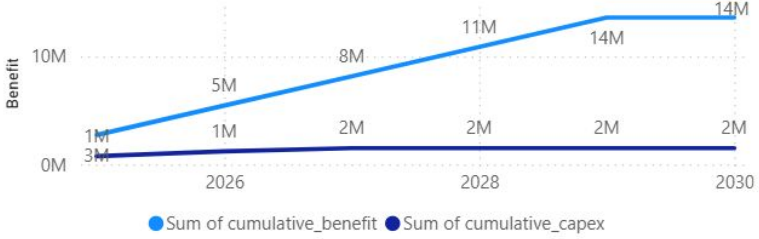
CAPEX Investment by Measure



ROI Analysis by Measure

Measure	Capex	Annual Benefit	Payback years
M1	300000	275520	1.00
M2	600000	54000	11.00
M3	200000	56000	3.00
Total	**1500000**	**854720**	**27.00**

Cumulative Investment vs Benefit (5-Year)



IMPLEMENTATION ROADMAP

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2024 (Q4 - NOW):

- ☐ Board approval for $1.5M investment
- ☐ Secure financing / budget allocation
- ☐ Begin supplier engagement program (M4)
- ☐ Lock in renewable energy PPAs (M1)

2025 (YEAR 1 - QUICK WINS):

- ☐ M1: Deploy renewable electricity → -3,444 tCO2e
- ☐ M3: Launch energy efficiency program → -700 tCO2e
- ✓ Expected reduction: 4,144 tCO2e (9% of baseline)
- ✓ Investment: $500K CAPEX
- ✓ Annual benefit: $331,520

2026 (YEAR 2):

- ☐ M2: Fleet electrification begins → -675 tCO2e
- ☐ M5: Logistics optimization → -195 tCO2e
- ✓ Cumulative reduction: 5,014 tCO2e (11% of baseline)
- ✓ Investment: $800K CAPEX
- ✓ Annual benefit: $401,520

2027 (YEAR 3 - CRITICAL):

- ☐ M4: Supplier engagement goes live → -5,670 tCO2e
- ✓ Cumulative reduction: 10,684 tCO2e (23% of baseline)
- ✓ Investment: $200K CAPEX (final)
- ✓ Annual benefit: $2.72M (total portfolio)
- ⚠️ CRITICAL: M4 is the key lever - must succeed for financial viability

2028-2030 (OPTIMIZATION PHASE):

- ☐ Optimize all measures performance
- ☐ Develop strategy for remaining 7,316 tCO2e gap
- ☐ Evaluate: offsets, product redesign, material substitution
- ☐ Monitor carbon pricing trajectory vs. projections

2030 (TARGET MILESTONE):

- ✓ Net Zero pathway achieved: 18,000 tCO2e
- ✓ vs BAU baseline: 52,000 tCO2e
- ✓ Reduction achieved: 34,000 tCO2e (74% below baseline)
- ✓ Remaining gap: 7,316 tCO2e (requires external action)
- ✓ Annual carbon cost saved: $2.72M
- ✓ Portfolio payback: 0.55 years achieved ✓

TOP 5 RISKS & MITIGATION

RISK #1: SUPPLIER PARTICIPATION (M4)

Severity: ⚠️ ⚠️ ⚠️
CRITICAL Problem: M4 = 53% of abatement. Need 40% supplier engagement by 2027.
Impact if fails: Loss of $453.6K annual benefit
Mitigation: Start negotiations NOW, offer incentives, phased approach

RISK #2: CARBON PRICING ASSUMPTION

Severity: ⚠️ ⚠️ MEDIUM
Problem: ROI assumes $80/tCO2e. @ $50/tCO2e benefit drops to $1.7M
Impact: ROI decreases from 81% to 13%
Mitigation: Lock in M1 renewable PPA at fixed rates, monitor policy
→ Pursue ESG benefits beyond financial (brand, compliance)

RISK #3: FLEET ELECTRIFICATION INFRASTRUCTURE (M2)

Severity: ⚠️ ⚠️ MEDIUM
Problem: M2 requires charging network. May not exist at scale.
Impact: M2 delayed, extended payback
Mitigation: Partner with charging networks NOW.

RISK #4: TECHNOLOGY READINESS

Severity: ⚠️ LOW
Problem: M1, M3 may underperform vs. guarantees
Impact: Reduced abatement, extended payback
Mitigation: Use proven tech only, performance guarantees, pilot M3

RISK #5: ECONOMIC DOWNTURN

Severity: ⚠️ ⚠️ MEDIUM
Problem: $1.5M CAPEX unaffordable if economy weakens
Impact: Project delays, missed target
Mitigation: Phased investment, prioritize high-ROI measures, green financing

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CONCLUSION

This Net Zero Transition Plan is **financially viable, strategically sound, and operationally achievable**, with clear ROI and manageable risks. The $1.5M investment generates $2.72M in annual savings, creating a compelling business case beyond ESG compliance.

Key Success Requirement: Immediate M4 supplier engagement (Q4 2024) and addressing the 69% remaining gap through product redesign, material substitution, or offsets.
The dashboard provides real-time visibility into progress and enables data-driven decision-making for the next 5 years.

CONTACT

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